

ROSS UNIVERSITY SCHOOL OF MEDICINE

DISORDERS OF THE HYPOTHALAMUS AND PITUITARY GLAND 2

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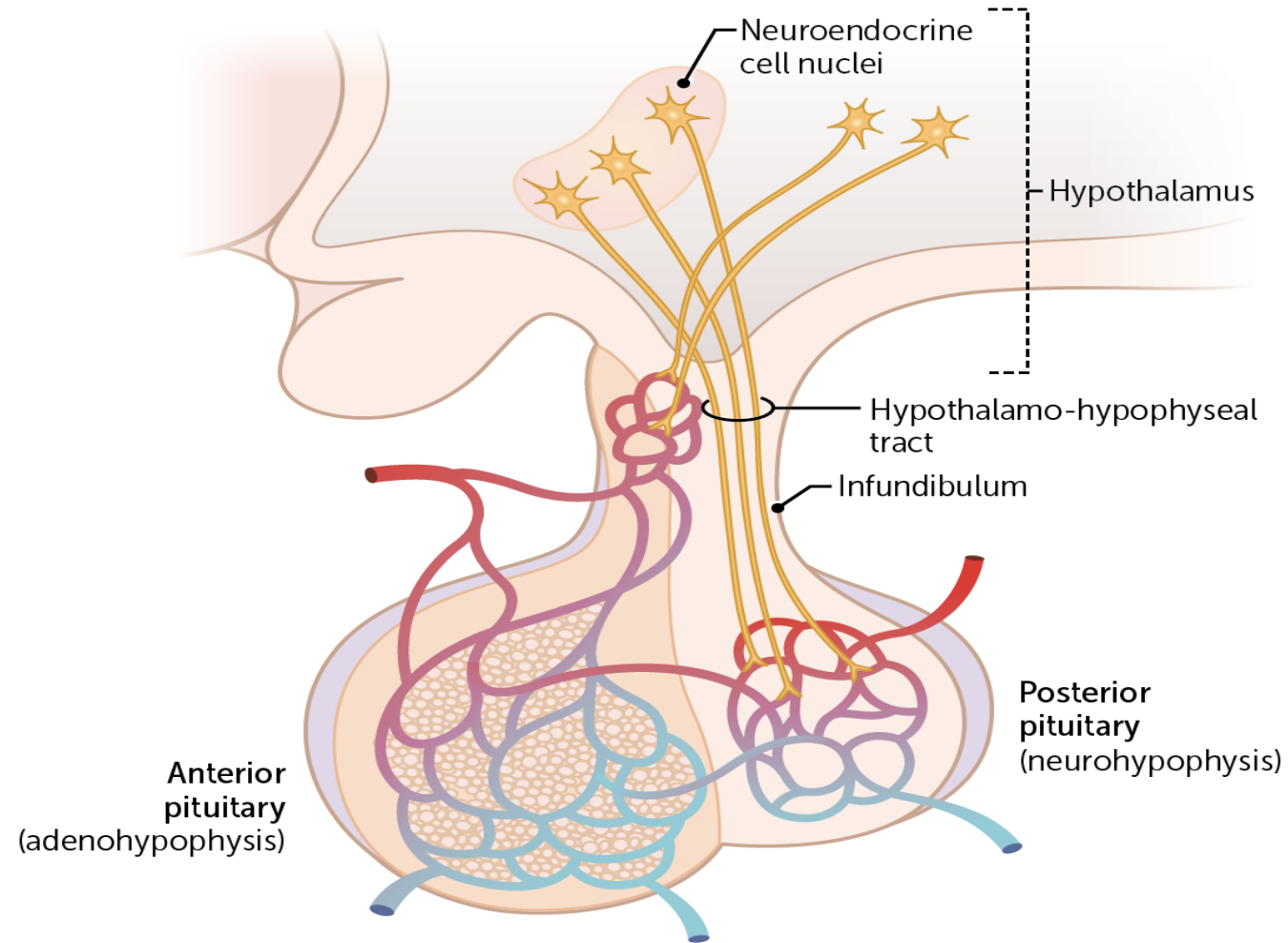
SCHOOL OF MEDICINE

OBJECTIVES



- Describe the etiology of pituitary dysfunction: Understand the underlying causes of pituitary dysfunction including underlying genetic factors, tumors, trauma, infections, and inflammatory conditions.
 - Pituitary tumors: Describe the clinical manifestations, diagnostic findings and management of pituitary tumors.
 - Hyperpituitarism: List the causes for hyperpituitarism, including acromegaly and Cushing's syndrome, and discuss the etiology, pathogenesis, morphology, investigations, and clinical features of pituitary hyperplasia, adenoma, and carcinoma.
 - Hypopituitarism: Describe the etiology, pathogenesis and clinical features of hypopituitarism and briefly discuss non-functional pituitary adenomas.
 - Posterior Pituitary Conditions: Identify the causes for posterior pituitary conditions and discuss the causes and significance of SIADH, including the importance of paraneoplastic syndrome.
 - Recall the embryology of Rathke pouch, explain the pathophysiology of craniopharyngioma, and describe its morphology, investigations, and clinical course.
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NORMAL HYPOTHALAMIC PITUITARY CONTROL





- Suspicion that the secretion of one or more is abnormal
- Signs or symptoms of hormone excess or deficiency
- Signs or symptoms suggestive of a deficiency in other pituitary hormones
- Follow-up for other abnormal hormone test results
- Monitor treatment
- Patients who have had radiation treatment of the central nervous system or whole-body irradiation



- Random of basal hormonal testing
- Stimulation test
 - *TRH*
 - *ACTH*
 - *GHRH*
 - *GnRH*
- Suppression test
 - *OGTT*
 - *Dexamethasone*

DISORDERS OF THE ENDOCRINE SYSTEM

Classification



- Hormonal excess causes
 - *Tumors*
 - *Hyperplasia*
 - *Autoimmune stimulation*
- Hormonal deficiency (80-90% of glandular tissue must be nonfunctional before obvious symptoms appear) causes
 - *Infection*
 - *Inflammation*
 - *Infarction*
 - *Infiltration*
 - *Hemorrhage*
 - *Autoimmunity*
- Hormonal resistance (Most cases involve water soluble hormones)
 - *Chronic high levels of hormone can induce tissue resistance by either:*
 - Receptor failure
 - Post-receptor intracellular signaling pathway problem

- Hypopituitarism
- Hormone deficiency
 - *Diabetes insipidus*
- Hormone excess
 - *Gigantism*
 - *Acromegaly*
 - *Hyperprolactinemia*
 - *SIADH*
 - *Cushing disease*
- Tumors
 - *Pituitary adenoma*
 - *Craniopharyngioma*

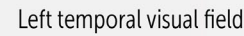
HYPOPITUARTISM

Caused by complete or partial loss of anterior pituitary hormones



- Major causes of hypopituitarism
 - *Hypothalamic disorders*
 - Mass lesions: benign (craniopharyngioma) and malignant tumors (mets from breast, lung etc)
 - Radiation: from CNS and nasopharyngeal malignancies
 - Infiltrative lesions: sarcoidosis and Langerhans cell histiocytosis
 - Infections: TB meningitis
 - Others: traumatic brain injury, strokes
 - *Pituitary disorders*
 - Mass lesions: pituitary adenomas, benign tumors, cysts
 - Pituitary surgery or radiation
 - Infiltrative lesions: hemochromatosis
 - Infection and abscesses
 - Infarction- Sheehan syndrome
 - Apoplexy
 - Genetic mutations and Empty sella

- Clinical presentation varies based on:
 - *How rapid the disease affects the anterior pituitary cells*
 - *The severity of the hormone deficiency*
 - *The number of different pituitary cells affected*
- Generally, GH> Gn> TSH> ACTH
- General symptoms: intrasellar/parasellar masses (e.g. pituitary macroadenoma, craniopharyngioma) can manifest with;
 - *Headache*
 - *Visual field defects (classic bitemporal hemianopsia)*
 - *And/or diplopia*



Hypopituitarism

Clinical Features



ACTH deficiency:

- weight loss
- weakness
- hypotension
- $\neq$ salt wasting
- volume contraction
- $\uparrow K^+$; $\downarrow Na^+$

TSH deficiency:

- fatigue
- cold intolerance
- weight gain
- decreased appetite
- constipation
- facial puffiness
- dry skin
- bradycardia
- delayed relaxation phase of the DTRs
- anemia

GH deficiency:

- ✓ Children
 - short stature
 - delayed bone age
 - decelerating growth velocity
 - hypoglycemia (neonates)
 - microphallus
 - delayed puberty
 - hair growth is sparse and thin
- ✓ Adults
 - usually asymptomatic
 - changes in body composition ($\uparrow$ fat mass, $\downarrow$ lean body mass)

FSH/LH deficiency:

- ✓ Women ($\downarrow$ estrogen secretion)
 - dis/amenorrhea
 - hot flashes
 - infertility
- ✓ Men ($\downarrow$ testosterone secretion)
 - $\downarrow$ libido
 - hot flashes
 - infertility
 - testicular atrophy
 - loss of facial, axillary and/or pubic hair
 - gynecomastia

Prolactin deficiency:

- ✓ Women: inability to lactate after delivery
- ✓ Men: asymptomatic

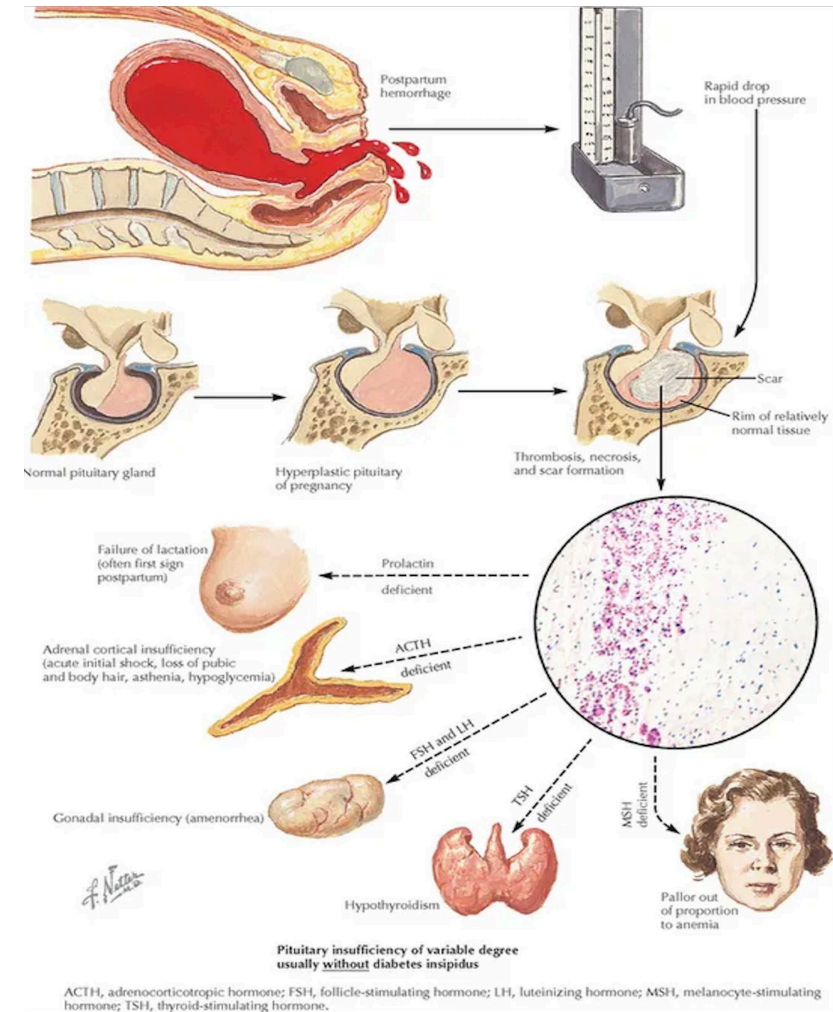


- Sudden onset of headache, visual symptoms, altered mental status, and hormonal dysfunction due to acute hemorrhage or infarction of a pituitary gland
- Signs and symptoms
 - *Acute severe headache*
 - *Nausea and vomiting*
 - *Changes in vision or visual fields, ptosis, diplopia*
 - *Sudden hypotension*
- Predisposing factors
 - *Head trauma*
 - *Pregnancy*
 - *Pituitary irradiation*
 - *Bromocriptine treatment*

SHEEHAN SYNDROME



- Results from a postpartum hemorrhage causing pituitary hypoperfusion
- Signs and symptoms
 - *Lactation failure*
 - *Amenorrhea*
 - *Decreased axillary and pubic hair*
 - *Signs of hypothyroidism and adrenal insufficiency*



Hypopituitarism

Diagnostic approach



- Test each hormone individually
 - *Pattern of hormone deficiency may vary.*
- If a hormone deficiency is identified next step is cranial imaging (preferably MRI) to identify the cause (e.g., pituitary adenoma).
- In an emergency setting (e.g., adrenal crisis, myxedema coma), treatment is indicated prior to biochemical testing

- Most common hormone secreting tumor in women of reproductive age
- Clinical features present earlier in females than males. Hence women often present with microadenomas and males with macroadenomas.
- Clinical features
 - *Visual defects in macroadenomas*
 - *In females*
 - oligomenorrhea, amenorrhea, galactorrhea, or infertility
 - *In males*
 - ↓libido, sexual dysfunction, field defects, headaches, features of hypoenestrogenism,
 - female body habitus and small testes: if develops prepubertally
 - Gynecomastia (rare)

- Pregnancy test
- Serum prolactin level
- Thyroid stimulating hormone (TSH)
 - *Thyrotropin releasing hormone (TRH) stimulates production of prolactin*
- BUN and creatinine
 - *Renal failure can cause hyperprolactinemia*
- Insulin like growth factor 1 (IGF-1)
 - *Due to risk of GH co-secretion in prolactinomas*
- Test other pituitary hormones to R/O hypopituitarism
- Visual field testing
- Radiological imaging: MRI preferred

PROLACTINOMA

Medical management



- More amenable to pharmacologic treatment than any other kind of pituitary adenoma
- Dopamine agonist is the treatment of choice: Bromocriptine or Cabergoline
 - *Treatment of symptoms*
 - *Reduce size*
- Treat any other endocrine abnormalities
- Regulate menses, desires pregnancy, impending neurosurgery due to the large size of the tumor
- Consider DEXA Scan in women with amenorrhea

PROLACTINOMA

Non-medical management



- Observation if the tumor is very small, patient is asymptomatic
- Surgical, Transsphenoidal Surgical Resection, for those patients that are:
 - *Intolerant or nonresponsive to medical therapy*
 - *Have progressive vision loss, rapid progression of the size of the tumor*
- Radiotherapy:
 - *For macroadenomas, non-surgical candidates or residual tumor post-surgery, aggressive or malignant tumors*

Craniopharyngioma

Clinical approach



- Slow growing tumor
- Peaks in 5-14 & 50-75 years
- Symptoms are based on the precise location & the adjacent structures to the tumor
 - *Visual symptoms due to optic nerve or chiasm compression*
 - *Endocrine abnormalities due to direct compression of normal structures.*
 - DI > Growth failure, TSH=ACTH deficiency
 - *Headaches*
 - *Other symptoms: depression, nausea, vomiting, lethargy*
- Radiology
 - *MRI and/or CT*
 - Calcification in the suprasellar region
 - Cystic calcified parasellar lesion





- Surgery
 - *Treatment of choice*
- Presurgical evaluation
 - *Endocrine testing especially of adrenal and thyroid function*
 - *Neuro ophthalmologic examination – visual field testing*
- Radiation
 - *Reserved for treatment of residual disease or recurrence*
- Long term complications:
 - *Panhypopituitarism*
 - *Second malignancies & tumor recurrence*
 - *Visual deficits*
 - *Neurologic*
 - *Hypothalamic dysfunction*

- Excess growth hormone (GH) causing increased insulin like growth factor-1 (IGF-1)
 - *Most often caused by pituitary tumor secreting GH*
 - *Rarely ectopic GH production or excess secretion of GHRH*
- Gigantism:
 - *Occurs during childhood while the epiphyseal plates are open*
- Acromegaly
 - *Occurs after the growth plates fuse in adulthood*

ACROMEGALY

Clinical features

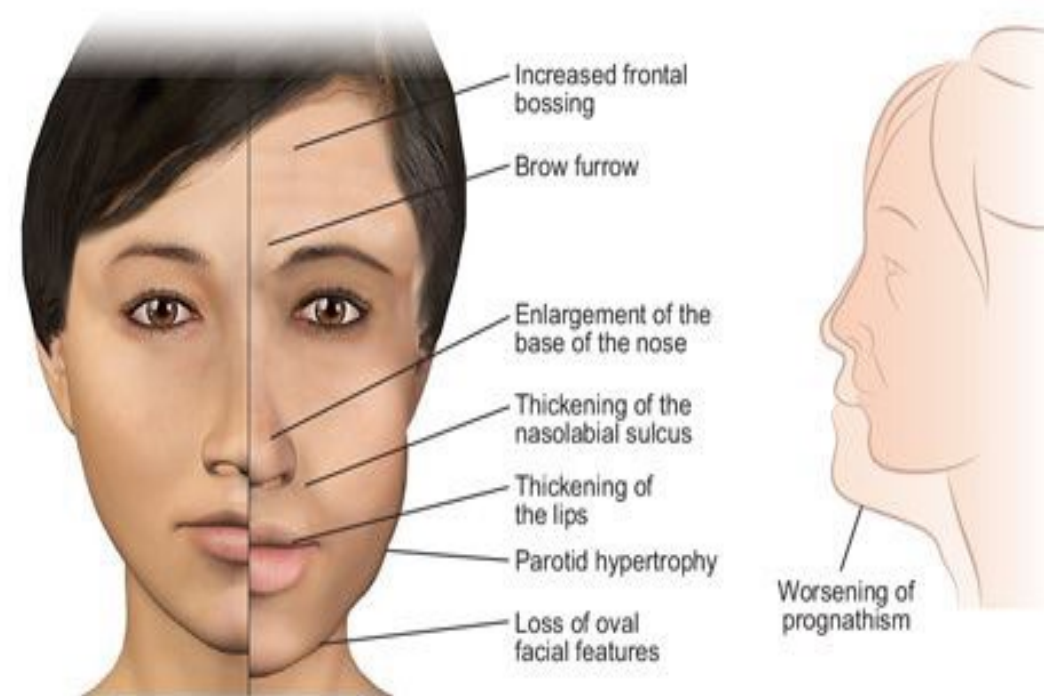
- Excess GH → stimulates hepatic secretion of IGF-1
- Diagnosis often delayed
- Symptoms depend on tumor size
 - *Enlargement during adulthood of the hands and feet -results in ↑ size of the shoes, rings and gloves*
 - *Coarse facial features, enlargement of the nose, frontal bones & jaw, (macroglossia) the upper incisors may spread apart, macroglossia*



ACROMEGALY

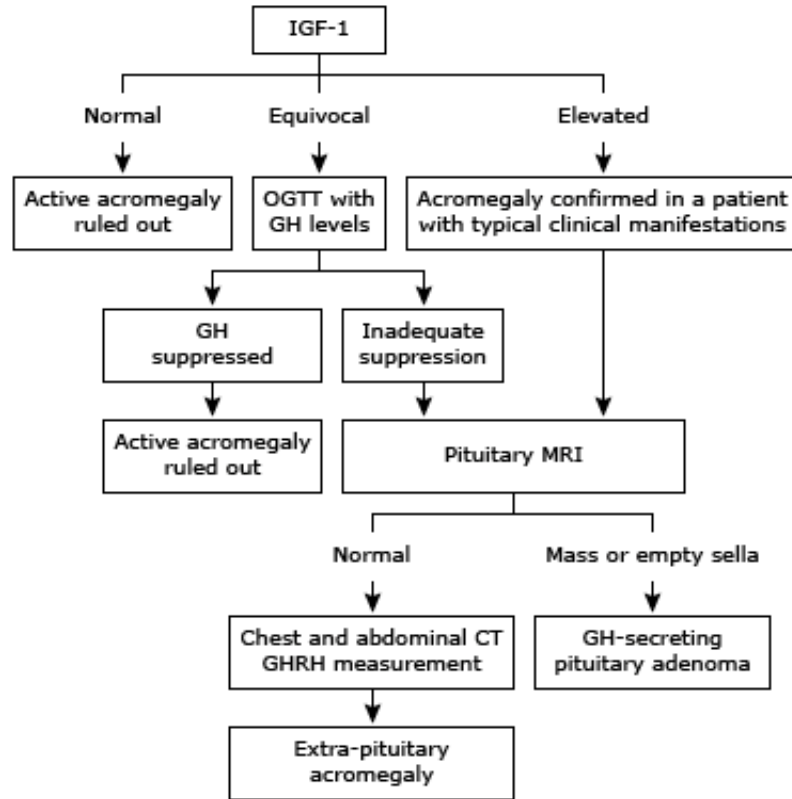
Clinical features

- ↑ incidence of colon polys & adenocarcinoma of the colon
- Sleep apnea
 - *due to macroglossia & enlargement of the soft tissue of the larynx & pharynx)*
- Hyperhidrosis, large skin pores, oily skin
- Arthritis, carpal tunnel syndrome
- HTN, CHF (cardiomyopathy)
- T2DM (hyperinsulinism, insulin resistance)



ACROMEGALY

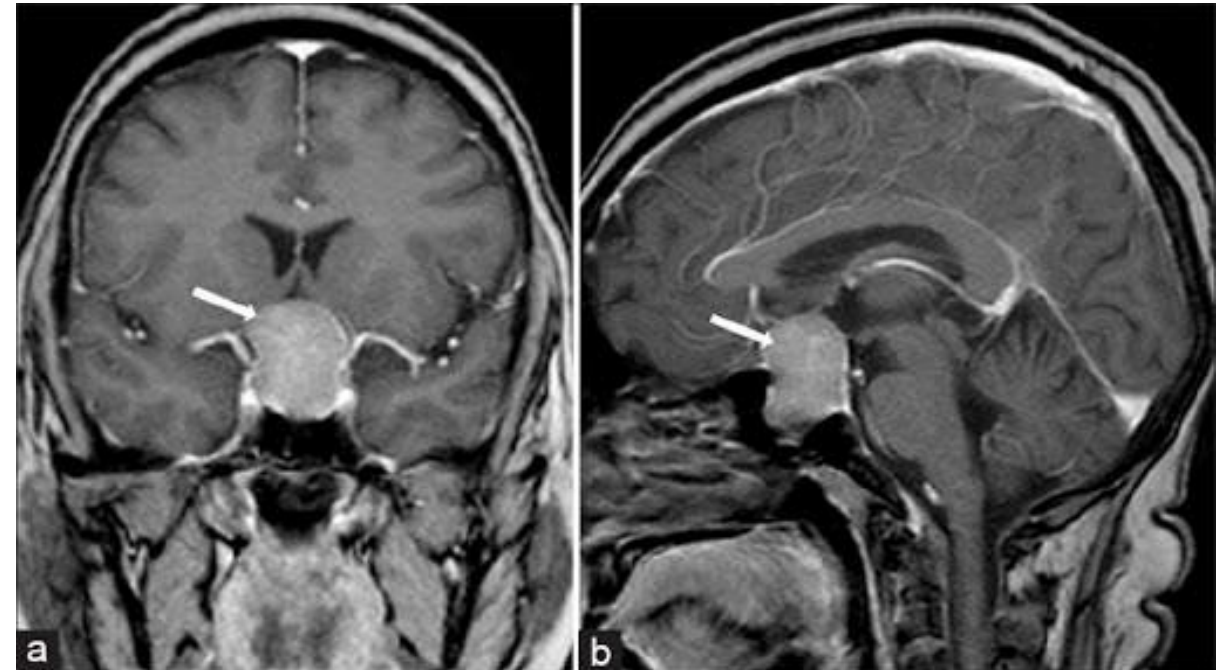
Algorithm for the Diagnosis



IGF-1: insulin-like growth factor-1; OGTT: oral glucose tolerance test; GH: growth hormone; MRI: magnetic resonance imaging; CT: computed tomography; GHRH: growth hormone-releasing hormone.

Adapted from: Melmed S. Anterior pituitary. In: Current Practice of Medicine, Korenman S (Ed), 1996.

UpToDate®



NB

MRI of the brain, ±CT Scan of the pancreas, adrenal or ovaries

□Radiography

□Demonstrate skeletal manifestations of GH/IGF-1 excess

GIGANTISM

Clinical features



- Presentation usually dramatic, organ enlargement including heart
- Rapid and accelerated linear growth
- Rapid GH precedes or is concurrent with rapid weight gain
 - *exogeneous obesity increase in weight before increase in height*
- Tumor mass effects
- Delayed puberty, irregular menstruation, galactorrhea
- Endocrine evaluation
 - *Random GH levels*
 - *Random IGF-1- most reliable biochemical indicator*
 - Indicator of endogenous GH secretion
 - Concentration is constant
 - Compare to the appropriate age matched range
 - *Assess for hypopituitarism*





- Surgery
 - *Transsphenoidal (treatment of choice)*
 - *Complications: CSF rhinorrhea, meningitis, recurrent or residual disease, transient DI*
- Goals of therapy
 - *Treat hormone excess and associated symptoms and comorbidities*
 - *Treat mass effects from the tumor*
 - *Assess for and treat any coexistent pituitary hormone deficiencies*
 - *Successful therapy:*
 - Normalizing IGF-1 levels for age
 - Maintaining appropriate GH levels
 - Preventing tumor mass effect on the optic nerves and other vital structures



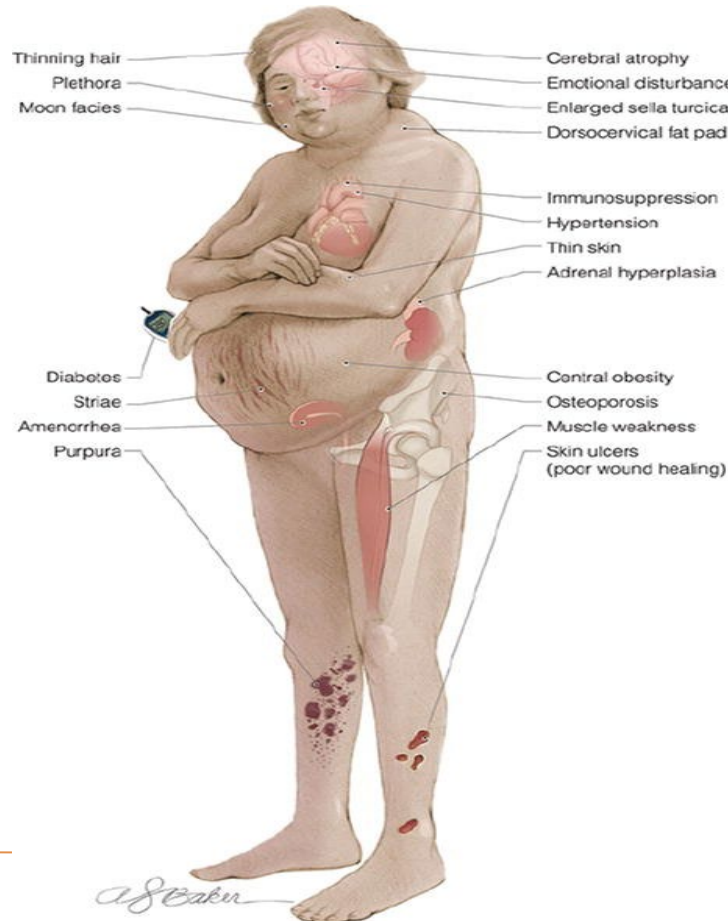
- Pharmacological

- *Use if incomplete surgical resection, patient refuses surgery, is not a surgical candidate*
- *Long acting somastatin analog* use a somatostatin analog first because it tends to decrease the size of the adenoma, as well as its secretion of GH or GH receptor antagonist (Pegvisomant)*
- *Combination therapy if monotherapy not successful: Long acting somastatin analog (Octreotide or Lanreotide) ± Pegvisomant*
- *Complications: based on medication used*

- Radiation

- *Useful adjunct therapy*
- *Complications: learning disability, obesity, emotional changes, hypopituitarism*

- Occurs because of ACTH hypersecretion (corticotroph adenoma) by the pituitary gland causing excess endogenous glucocorticoid production.
- Corticotroph adenomas usually cause Cushing's disease, but a significant minority are "clinically silent"
- Symptoms:
 - *Weight gain*
 - *Thin skin*
 - *Striae*
 - *Proximal muscle weakness*
 - *HTN*
 - *Osteoporosis*





- Confirm excess cortisol
 - *Late-night salivary cortisol X2 (<1.8 mcg/dl = normal)*
 - *24-hour urinary free cortisol ($>3X$ upper limit of normal)*
 - *Or overnight dexamethasone suppression test*
- MRI of the brain
- Inferior petrosal sinus/venous sampling (IPSS):
 - *Measure ACTH gradients between two sinuses during inferior sinus catheterization to localize the tumor*



- Transsphenoidal surgery
 - *Treatment of choice*
- Pituitary radiotherapy
 - *Complication: hypopituitarism*
- Medical therapy :
 - *Use in patients that are;*
 - Not surgical candidates
 - Do not achieve remission after surgery
 - *Adrenal enzyme inhibitors are the most commonly used drugs*
 - *Adrenolytic agents, drugs that target the pituitary, and glucocorticoid-receptor antagonists are also used*

CENTRAL DIABETES INSIPIDUS (DI)

Due to Vasopressin Deficiency



- Results from deficiency of ADH (vasopressin)
- Passage of $> 3\text{L}$ in 24 hours of dilute urine ($<300\text{ mOsm/kg}$)
- Causes
 - *Idiopathic accounts for 30-50% of cases*
 - *Primary or secondary tumors*
 - *Infiltrative disease*
 - *Neurosurgery*
 - *Trauma*
 - *Associated with familial and congenital diseases (septo-optic dysplasia, Wolfram syndrome)*

CENTRAL DI

Differential diagnoses of diabetes insipidus

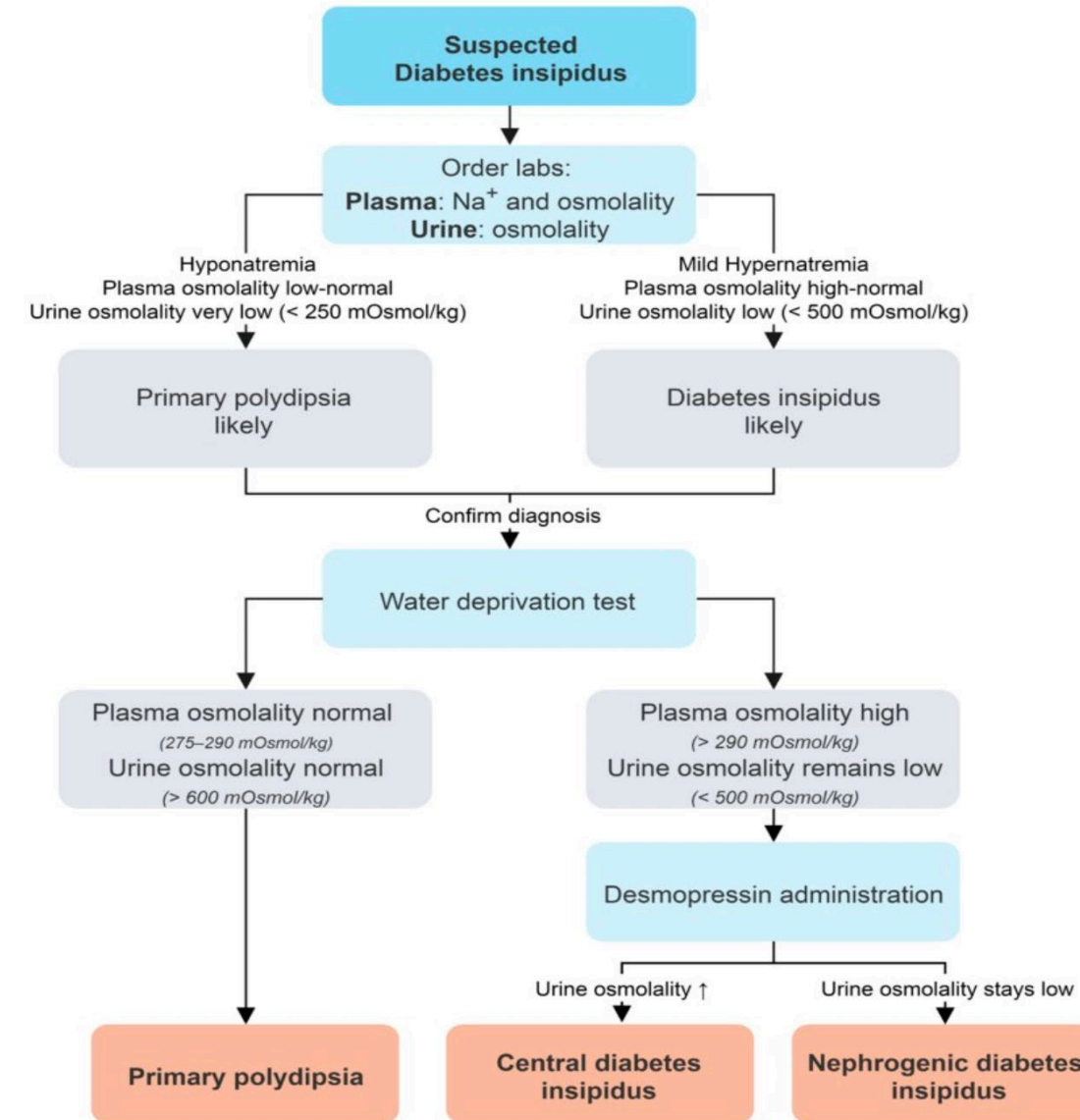


- Nephrogenic DI
- Langerhans Cell Histiocytosis
- Medullary Cystic Disease
- Hypercalcemia
- Hypokalemia
- Head trauma
- Type 1 DM
- Sickle cell anemia



- Polyuria, nocturia
- Dehydration, polydipsia
- Babies
 - *Irritability, crying*
 - *Thin, weight loss, failure to thrive*
 - *Hyperthermia*
- Children
 - *Anorexia, excessive fatigue, enuresis*
 - *Linear growth defects*
- Physical examination varies with severity and chronicity of DI
 - *Orthostatic hypotension, ↓ BP, ↑ HR*
 - *Dry mucous membranes, skin tenting*
 - *± bladder enlargement*

- History: rate of onset of polyuria
- Confirm polyuria with 24 hr urine collection for vol
 - $> 3 \text{ L/d}$ in adults;
 - $> 2 \text{ L/m}^2/\text{d}$ children with 24
- Urine osmolarity (Uosm) $< 300 \text{ mOsm/kg}$
- Baseline serum
 - electrolytes
 - glucose
 - calcium
 - osmolality
- Water deprivation test





- Begin the night, withhold all water intake
- Baseline then hourly measurement of body weight & Uosm
- When 2 sequential Uosm vary < 30 mOsm/kg or body weight $\downarrow > 3\%$ GIVE 5U aqueous ADH or desmopressin subcutaneously
- Recheck Uosm 60 minutes post Water Deprivation Test
- Results in healthy individual
 - $Uosm\ 2-4 \times Posm$ (plasma osmolarity)
 - ADH administration produces $\uparrow < 9\%$ Uosm
- Results in Central/Nephrogenic DI
 - $Uosm < 300$ mOsm/kg
 - Post desmopressin in:
 - Central DI: $Uosm > 750$ mOsm/kg
 - Nephrogenic DI: Uosm no change
 - Primary polydipsia: $Uosm > 750$ mOsm/kg

- Goals
 - *Prevention of dehydration*
 - *Reduce nocturia, thereby providing adequate sleep*
 - *Aim for **partial** control of the diuresis during the day*
 - *Prevent hyponatremia*
- ADH (vasopressin) synthetic analogues
 - ***Desmopressin** initially administer at night*
 - *Drug of choice*
- Nonhormonal drugs
 - *Rarely used – have any diuretic properties*
 - *example: chlorpromide, carbamazepine, thiazide diuretics, and nonsteroidal anti-inflammatory drugs (NSAIDs)*

- Caused by the inability to suppress the secretion of ADH
- Disorder of impaired water excretion
- Defined by the hyponatremia and hypo-osmolality
- Water intake exceeds reduced urine output → water retention → hyponatremia
- Etiology:
 - *Any CNS disorder, including stroke, hemorrhage, infection, trauma, and psychosis*
 - *Ectopic ADH production by small cell carcinoma of the lung*
 - *Pulmonary disease (pneumonia, asthma, ARDS)*
 - *Drugs that enhance ADH release or effect:*
 - Chlorpropamide, carbamazepine, cyclophosphamide
 - selective serotonin reuptake inhibitors
 - *Neurosurgical procedures*



- Asymptomatic
- Mild/moderate symptoms
 - *Dizziness*
 - *Gait disturbance*
 - *Forgetfulness*
 - *Lethargy*
 - *Headaches*
 - *Nausea, vomiting*
- Severe symptoms ($\text{Na} < 120 \text{ mEq/L}$)
 - *Seizures*
 - *inability to communicate &/or coma*



- SIADH is a diagnosis of exclusion, and therefore, other causes of hyponatremia must be ruled out
- Low plasma osmolality: $< 280 \text{ mOsm/kg}$
- Inappropriately elevated:
 - *Urine osmolality* $> 100 \text{ mOsm/kg}$
 - *Urine sodium concentration* $> 40 \text{ mEq/L}$
- Euvolemia: volume depletion or overload should prompt an evaluation for an alternative Dx. SIADH should be considered if hypo-osmolar hyponatremia persists once clinical euvolemia is restored.
- Normal serum potassium concentration



- Treatment of the underlying cause
- Avoid rapid correction of the serum sodium
 - *Central pontine myelinolysis*
- Initial therapy is to raise the serum sodium
- Prolonged therapy in patients with persistent SIADH
- Fluid restriction is mainstay of therapy
 - $< 800\text{mls/day}$
- Hypertonic 3% NaCl or oral salt tablets
 - *If symptomatic*
 - *Seizures*
- Oral salt tablets and loop diuretics
- Chronic SIADH
 - *Vasopressin receptor antagonist*